Example 8.8

Calculate i) degree of hydrolysis, ii) the hydrolysis constant and iii) pH of 0.1M CH_3COONa solution (pK_a for CH_3COOH is 4.74).

Solution (a) CH_3COONa is a salt of weak acid (CH_3COOH) and a strong base (NaOH). Hence, the solutions is alkaline due to hydrolysis.

$$CH_3COO^-(aq) + H_2O(aq) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$$

i) $h = \sqrt{\frac{K_w}{K_a \times C}}$

$= \sqrt{\frac{1 \times 10^{-14}}{1.8 \times 10^{-5} \times 0.1}}$

$h = 7.5 \times 10^{-5}$

Give that $pK_a = 4.74$

$pK_a = -\log K_a$

i.e., K_a = antilog of $(-pK_a)$

= antilog of (-4.74)

= antilog of (-5 + 0.26)

$= 10^{-5} \times 1.8$

[antilog of 0.26 = 1.82 $\simeq$ 1.8]

ii) $K_h = \frac{K_w}{K_a} = \frac{1 \times 10^{-14}}{1.8 \times 10^{-5}}$

$= 5.56 \times 10^{-10}$

iii) $pH = 7 + \frac{pK_a}{2} + \frac{\log C}{2}$

$= 7 + \frac{4.74}{2} + \frac{\log 0.1}{2} = 7 + 2.37 - 0.5$

$= 8.87$

Evaluate yourself - 10

Calculate the i) hydrolysis constant, ii) degree of hydrolysis and iii) pH of 0.05M sodium carbonate solution (pK_a for HCO_3^- is 10.26).

8.9 Solubility Product

We have come across many precipitation reactions in inorganic qualitative analysis. For example, dil HCl is used to precipitate Pb^{2+} ions as $PbCl_2$ which is sparingly soluble in water. Kidney stones are developed over a period of time due to the precipitation of Ca^{2+} (as calcium oxalate etc…). To understand the precipitation, let us consider the solubility equilibria that exist between the undissociated sparingly soluble salt and its constituent ions in solution.

For a general salt X_mY_n,

$$X_mY_n(s) \overset{H_2O}{\rightleftharpoons} mX^{n+}(aq) + nY^{m-}(aq)$$

The equilibrium constant for the above is

$$K = \frac{[X^{n+}]^m [Y^{m-}]^n}{[X_mY_n]}$$

In solubility equilibria, the equilibrium constant is referred as solubility product constant (or) Solubility product.

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